

04-src-foundation

04 — src foundation extraction

Source: docs/research/browser_extension/Browser Torrent Extension Guide/src/ Reference browser extension **BobaLink** (WXT + TypeScript + Manifest V3). This is the foundation layer (build/config + data layer + shared libs + types). Every file below was read IN FULL. Exact values are quoted with file + line citations.

Files read

File	Lines
package.json	61
tsconfig.json	37
wxt.config.ts	169
jest.config.ts	143
playwright.config.ts	133
.eslintrc.json	56
.prettierrc	14
_locales/en/messages.json	118
.github/workflows/ci.yml	55
.github/workflows/release.yml	416
sql/schemas.sql	276
sql/migrations/001_initial.sql	267
src/types/api.ts	486
src/types/config.ts	232
src/types/torrent.ts	223
src/shared/constants.ts	403
src/shared/crypto.ts	292
src/shared/errors.ts	342

File	Lines
src/shared/events.ts	175
src/shared/logger.ts	167
src/shared/storage.ts	248
src/shared/utils.ts	366

1. Package metadata, dependencies, npm scripts (package.json)

- **name** bobalink, **version** 1.0.0, **type** module, **license** MIT, **author** Boba Project.
- **engines.node** >=20.0.0.
- **repo** <https://github.com/boba-project/bobalink.git> (HTTPS — note Boba constitution mandates SSH; placeholder only).
- **description**: “Browser extension that detects torrent files and magnet links, sending them to Boba Project’s qBitTorrent dashboard”.

Dependencies (1)

Package	Version
webextension-polyfill	^0.12.0

devDependencies (15)

Package	Version
@playwright/test	^1.49.0
@types/chrome	^0.0.287
@types/jest	^29.5.14
@types/node	^22.10.0
@types/webextension-polyfill	^0.12.1
@typescript-eslint/eslint-plugin	^8.18.0
@typescript-eslint/parser	^8.18.0
@wxt-dev/auto-icons	^1.0.2
eslint	^9.17.0
jest	^29.7.0
jest-environment-jsdom	^29.7.0
prettier	^3.4.0
ts-jest	^29.2.0
ts-node	^10.9.2

Package	Version
typescript	^5.7.0
wxt	^0.19.0

npm scripts (all 15)

Script	Command
dev	wxt
dev:firefox	wxt -b firefox
build	wxt build
build:firefox	wxt build -b firefox
zip	wxt zip
zip:firefox	wxt zip -b firefox
compile	tsc --noEmit
lint	eslint src/ --ext .ts
lint:fix	eslint src/ --ext .ts --fix
format	prettier --write "src/**/*.ts"
format:check	prettier --check "src/**/*.ts"
test	jest --coverage
test:watch	jest --watch
test:e2e	playwright test
test:e2e:ui	playwright test --ui
postinstall	wxt prepare

Note: package.json references wxt build (chrome default) + build:firefox. The **release.yml** workflow references build:chrome / build:firefox scripts that **do not exist** in package.json (drift in the reference repo — flagged for the plan).

2. Manifest / permissions / CSP config (wxt.config.ts + _locales/en/messages.json)

WXT generates manifest.json from wxt.config.ts. Extracted manifest config:

- **manifest_version:** 3
- **name:** __MSG_extName__ → “BobaLink”
- **description:** __MSG_extDescription__
- **version:** 1.0.0
- **default_locale:** en

- **minimum_chrome_version:** "109"
- **outDir:** dist

Permissions (6)

storage, alarms, notifications, activeTab, scripting, contextMenus

host_permissions (3) — the load-bearing endpoint set

- http://localhost:7187/* — Boba FastAPI server
- http://localhost:7189/* — Boba Go server
- http://localhost:8080/* — qBitTorrent WebUI

Mismatch vs the real Boba port map: actual qBitTorrent WebUI proxy is **7186** (CLAUDE.md port table), merge service **7187**, boba-jackett **7189**. The extension hardcodes **8080** for qBitTorrent and assumes FastAPI on 7187 / Go on 7189. The plan must reconcile (qBittorrent proxy is 7186, not 8080).

action (toolbar popup)

- default_popup: popup/index.html
- default_icon: 16 → icon-16.png, 32 → icon-32.png
- default_title: __MSG_extName__

background (service worker)

- service_worker: background.js
- type: module (ES module service worker)

content_scripts (1 entry)

- matches: ["<all_urls>"]
- js: ["content-scripts/content.js"]
- css: ["content-scripts/content.css"]
- run_at: document_idle

options_page

- options/index.html

web_accessible_resources

- resources ["assets/*"], matches ["<all_urls>"]

content_security_policy (extension_pages)

```
default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'; connect-src 'self' http://localhost:7187 http://localhost:7189 http://localhost:8080; img-src 'self' data;;
```

- Strict: no inline scripts (script-src 'self'); style-src allows 'unsafe-inline'; connect-src restricted to the three localhost backends; img-src allows data: URIs.

commands (keyboard shortcuts, 3)

Command	Default	Mac	Description (msg)
send-to-boba	Ctrl+Shift+B	Command+Shift+B	__MSG_cmdSendToBoba__
scan-page	Ctrl+Shift+S	Command+Shift+S	__MSG_cmdScanPage__
open-dashboard	Ctrl+Shift+D	Command+Shift+D	__MSG_cmdOpenDashboard__

WXT build options

- autoIcons.baseIconPath: src/assets/icon.svg (auto-generates icon sizes from one SVG)
- dev.port: 3000
- runner.disabled: true (no auto browser launch)

i18n messages (_locales/en/messages.json) — 28 keys

Key UI strings: extName=“BobaLink”, extDescription, command descriptions (cmdSendToBoba/cmdScanPage/cmdOpenDashboard), popup (popupTitle=“BobaLink - Detected Torrents”, noTorrentsDetected, sendToBoba, scanning, sending, sendSuccess, sendFailed), connection states (connectionOnline=“Connected”, connectionOffline=“Disconnected”, notConfigured=“No server configured”), options page sections (optionsTitle, serverSectionTitle, generalSectionTitle, advancedSectionTitle), buttons (addServer, testConnection, autoDiscover, saveChanges, discardChanges), type labels (magnetLink, torrentFile), context menu (contextMenuSend, contextMenuScan, contextMenuDashboard).

3. TypeScript compiler config (tsconfig.json)

Option	Value
target	ES2022

Option	Value
module	ESNext
moduleResolution	bundler
lib	["ES2022", "DOM", "DOM.Iterable"]
jsx	preserve
strict	true
noImplicitAny	true
strictNullChecks	true
strictFunctionTypes	true
strictBindCallApply	true
strictPropertyInitialization	true
noImplicitReturns	true
noFallthroughCasesInSwitch	true
noUncheckedIndexedAccess	true
exactOptionalPropertyTypes	true
noUnusedLocals	true
noUnusedParameters	true
declaration	true
declarationMap	true
sourceMap	true
esModuleInterop	true
skipLibCheck	true
forceConsistentCasingInFileNames	true
resolveJsonModule	true
isolatedModules	true
types	["chrome", "jest", "node", "webextension-polyfill"]
baseUrl	.
paths	~/* → src/*, @/* → src/*
include	src/**/*.ts, tests/**/*.ts, wxt.config.ts, jest.config.ts, playwright.config.ts
exclude	node_modules, dist, .wxt

Very strict TS posture (all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes).

ESLint (.eslintrc.json)

- parser: @typescript-eslint/parser; project: ./tsconfig.json (type-aware lint).

- extends: eslint:recommended, plugin:@typescript-eslint/recommended, ...recommended-requiring-type-checking, ...strict.
- env: browser, es2022, node, webextensions, jest.
- Notable rules: no-explicit-any **error**; no-floating-promises **error**; no-misused-promises **error**; prefer-nullish-coalescing **error**; no-console warn (allows error/warn/info); eeqeq error (null-ignore); no-unused-vars error with ^_ ignore. tests/** override turns off unsafe/any rules. Ignores dist/, .wxt/, node_modules/, coverage/.

Prettier (.prettierrc)

```
semi:true, trailingComma:"all", singleQuote:false, printWidth:100,
tabWidth:2, useTabs:false, bracketSpacing:true, arrowParens:"always",
endOfLine:"lf", quoteProps:"as-needed", bracketSameLine:false,
proseWrap:"preserve".
```

Jest (jest.config.ts)

- preset ts-jest/presets/default-esm, testEnvironment: jsdom, ESM (extensionsToTreatAsEsm: [".ts"]).
- roots tests/unit + src; testMatch **/tests/unit/**/*test.ts, **/src/**/*test.ts.
- moduleNameMapper mirrors tsconfig paths (~/, @/).
- **coverageThreshold global 80% (branches/functions/lines/statements).**
- collectCoverageFrom: src/**/*ts excluding *.d.ts, index.ts, assets/, popup/, options/, content/, background/ (i.e. covers logic libs only — parsers/api/scanner/shared).
- setupFiles tests/unit/chrome-mock.ts; setupFilesAfterEnv tests/unit/setup.ts; clearMocks+restoreMocks; testTimeout: 10000; reporters text, text-summary, lcov, html → coverage/.

Playwright (playwright.config.ts)

- testDir ./tests/e2e; fullyParallel: false; workers: 1 (extension state); retries 2 in CI; reporters html+list (+github in CI).
 - use.baseURL chrome-extension://test-id/; trace on-first-retry; screenshot only-on-failure; video on-first-retry; viewport 1280×720.
 - launchOptions load unpacked ext via --load-extension=\${EXTENSION_PATH}||./dist}.
 - projects: chromium (channel chromium) + firefox.
 - webServer: command: npm run build, url http://localhost:3000, timeout 120000.
 - globalSetup/globalTeardown: ./tests/e2e/global-setup.ts / global-teardown.ts (**these files do not exist in the tree** — referenced only).
 - timeout: 30000, expect timeout 5000.
-

4. Data model — SQL schema + TypeScript types

4a. SQLite schema (sql/schemas.sql = migrations/001_initial.sql, schema v1.0.0)

Persisted via **chrome.storage.local** with an **sql.js** wrapper (header comment). 9 tables, ~25 indexes (header says “7” but actual index count is higher), 30 default site selectors, 25 default config rows. Migration file wraps everything in BEGIN TRANSACTION; ... COMMIT; and adds a migration_id='001_initial' metadata row.

Table 1 — app_metadata (key/value app state) | Col | Type | Constraint | |---|---|---| | key | TEXT | PRIMARY KEY | | value | TEXT | NOT NULL | | updated_at | INTEGER | NOT NULL |

Seeded keys: schema_version='1.0.0', app_name='BobaLink', app_version='1.0.0', installed_at, last_db_vacuum (migration also seeds migration_id='001_initial').

Table 2 — extension_config (typed key-value settings) | Col | Type | Constraint | |---|---|---| | key | TEXT | PRIMARY KEY | | value | TEXT | NOT NULL | | data_type | TEXT | CHECK IN (string,number,boolean,json,encrypted) NOT NULL | | description | TEXT | | updated_at | INTEGER | NOT NULL |

25 seeded config rows (key, value, data_type): -

server.base_url=http://localhost:8080 (string),

server.auth_method=cookie (string),

server.api_key=(**encrypted**), `server.username`= (string),

server.password=(**encrypted**) - `download.default\_category`= ,

download.default_save_path=`,download.start_paused=false(bool),download.auto_n

-behavior.auto_scan=true,behavior.highlight_detected=true,behavior.show_notifica

-security.require_https=true,security.encrypt_credentials=true,security.session_tir

-ui.theme=auto,ui.popup_width=400,ui.popup_height=600,ui.show_tracker_list=tr

Table 3 — discovered_torrents (per-session discovery) | Col | Type | Notes | |---|---|---| | id | TEXT | PRIMARY KEY — infoHash (40-char hex) | | page_url | TEXT | NOT NULL | | page_title | TEXT | | magnet_uri | TEXT | | torrent_url | TEXT | | torrent_data_b64 | TEXT | Base64 .torrent content | | name | TEXT | | display_name | TEXT | URL-decoded for UI | | trackers | TEXT | JSON array | | size_bytes | INTEGER | nullable | | source_type | TEXT | CHECK IN (magnet-link,torrent-file,torrent-url,infohash) NOT NULL | | discovery_method | TEXT | CHECK IN (link-scan,text-scan,site-specific>manual) DEFAULT link-scan | | is_private | INTEGER | DEFAULT 0 | | detected_at | INTEGER | NOT NULL | | sent_to_boba | INTEGER | DEFAULT 0 | | sent_at | INTEGER | | boba_status | TEXT | CHECK IN (pending,queued,sending,added,duplicate,error,retrying) DEFAULT

pending | | boba_error | TEXT | | | tab_id | INTEGER | | |
tab_group_id | INTEGER | | | selected | INTEGER | DEFAULT 1 | |
metadata_json | TEXT | |

Table 4 — download_queue (offline support) | Col | Type | Notes | |
---|---|---| | id | INTEGER | PK AUTOINCREMENT | | info_hash
| TEXT | NOT NULL | | magnet_uri | TEXT | | | torrent_data_b64 |
TEXT | | | name | TEXT | | | category | TEXT | | | save_path | TEXT |
| | tags | TEXT | JSON array | | add_paused | INTEGER | DEFAULT
0 | | skip_checking | INTEGER | DEFAULT 0 | |
sequential_download | INTEGER | DEFAULT 0 | |
first_last_piece_prio | INTEGER | DEFAULT 0 | | added_at |
INTEGER | NOT NULL | | retry_count | INTEGER | DEFAULT 0 | |
last_retry_at | INTEGER | | | next_retry_at | INTEGER |
exponential backoff | | last_error | TEXT | | | error_code | TEXT | | |
status | TEXT | CHECK IN
(pending, retrying, failed_permanent, completed, cancelled) DEFAULT
pending | | server_url | TEXT | NOT NULL | | server_id | INTEGER |
→ server_config.id |

Table 5 — server_config (multi-server) | Col | Type | Notes | |---|
---|---| | id | INTEGER | PK AUTOINCREMENT | | name | TEXT
| NOT NULL | | base_url | TEXT | NOT NULL **UNIQUE** | |
auth_method | TEXT | CHECK IN (none, cookie, api_key, basic) NOT
NULL DEFAULT none | | api_key_encrypted | TEXT | | | username |
TEXT | | | password_encrypted | TEXT | | | is_default | INTEGER |
DEFAULT 0 | | is_reachable | INTEGER | DEFAULT 0 | |
last_check_at | INTEGER | | | last_check_result | TEXT | CHECK IN
(success, auth_required, unreachable, error) | | qbittorrent_version |
TEXT | | | boba_version | TEXT | | | boba_fastapi_port | INTEGER |
DEFAULT **7187** | | boba_go_port | INTEGER | DEFAULT **7189** | |
qbittorrent_port | INTEGER | DEFAULT **8080** | | created_at |
INTEGER | NOT NULL | | updated_at | INTEGER | NOT NULL | |
notes | TEXT | |

Table 6 — send_history (immutable send log) | Col | Type | Notes |
|---|---|---| | id | INTEGER | PK AUTOINCREMENT | |
info_hash | TEXT | NOT NULL | | name | TEXT | | | magnet_uri |
TEXT | | | source_type | TEXT | NOT NULL | | sent_at | INTEGER |
NOT NULL | | server_id | INTEGER | REFERENCES
server_config(id) | | server_url | TEXT | NOT NULL | | success |
INTEGER | NOT NULL (1/0) | | response_code | INTEGER | HTTP
code | | error_message | TEXT | | | torrent_added_hash | TEXT |
hash returned by qBitTorrent |

Table 7 — site_selectors (CSS selectors per site) | Col | Type |
Notes | |---|---|---| | id | INTEGER | PK AUTOINCREMENT | |
domain_pattern | TEXT | NOT NULL (e.g. thepiratebay.*) | |
selector_type | TEXT | CHECK IN (magnet-link, torrent-
link, name, size, seeders, leechers, date, uploader) NOT NULL | |
css_selector | TEXT | NOT NULL | | attribute | TEXT | e.g. href,

null for text || regex_filter | TEXT | optional || priority | INTEGER
| DEFAULT 100 (lower = higher prio) || is_enabled | INTEGER |
DEFAULT 1 || notes | TEXT ||

30 seeded selector rows covering: thepiratebay, 1337x
(magnet+torrent), nyaa.si (magnet+torrent), rutracker
(magnet+torrent), kinozal, nnm-club, iptorrents (torrent),
limetorrents, torrentgalaxy, glodls, torlock, eztv, yts, rarbg,
demonoid, torrentfunk, yourbittorrent, bitsearch, bt4g,
solidtorrents, knaben, snowfl, bittorrent, megapeer,
badasstorrents, extratorrent. Most use a[href^="magnet:"] @
priority 10; direct .torrent links @ priority 20 (or 10 for
iptorrents).

Table 8 — queue_log (queue event log) | Col | Type | Notes | |—|
—|—|—| | id | INTEGER | PK AUTOINCREMENT | |
queue_item_id | INTEGER | REFERENCES download_queue(id) | |
action | TEXT | NOT NULL CHECK IN
(enqueue, retry, success, fail, cancel, skip) || timestamp | INTEGER |
NOT NULL || details | TEXT | JSON || error | TEXT ||

Table 9 — statistics (daily aggregates) | Col | Type | Notes | |—|
—|—|—| | id | INTEGER | PK AUTOINCREMENT | | stat_type |
TEXT | NOT NULL CHECK IN
(torrents_detected, torrents_sent, torrents_failed, api_calls, api_errors, queue_items, qu
|| stat_value | INTEGER | NOT NULL DEFAULT 0 || stat_date |
TEXT | NOT NULL (YYYY-MM-DD) || created_at | INTEGER | NOT
NULL || — | — | **UNIQUE(stat_type, stat_date)** |

Indexes (CREATE INDEX IF NOT EXISTS): - discovered_torrents:
idx_discovered_page_url(page_url),
idx_discovered_status(boba_status), idx_discovered_tab(tab_id),
idx_discovered_detected(detected_at),
idx_discovered_selected(selected) WHERE selected=1 (partial) -
download_queue: idx_queue_status(status),
idx_queue_next_retry(next_retry_at), idx_queue_hash(info_hash),
idx_queue_server(server_url) - send_history:
idx_history_hash(info_hash), idx_history_date(sent_at),
idx_history_server(server_id) - extension_config:
idx_config_key(key) - server_config:
idx_server_default(is_default) WHERE is_default=1 (partial),
idx_server_url(base_url) - site_selectors:
idx_selectors_domain(domain_pattern),
idx_selectors_enabled(is_enabled) WHERE is_enabled=1 (partial) -
queue_log: idx_queue_log_item(queue_item_id),
idx_queue_log_timestamp(timestamp) - statistics:
idx_stats_type_date(stat_type, stat_date)

Compatible with SQLite 3.38+ / sql.js. **§11.4.10 note:** extension_config stores server.api_key and server.password as data_type='encrypted'; server_config has api_key_encrypted/password_encrypted columns — credentials are designed to be encrypted-at-rest (see §6 crypto).

4b. TypeScript domain types

src/types/torrent.ts

- **interface MagnetInfo:** uri:string, infohash:string (40-char hex SHA-1), displayName:string|null, trackers:readonly string[], webSeeds:readonly string[], exactLength:number|null, exactSource:string|null, keywords:readonly string[], acceptableSource:string|null, manifest:string|null, detectedAt:number, sourceElement:Element|null. (Covers magnet params xt/dn/tr/ws/xl/xs/kt/as/mt.)
- **interface TorrentFile:** url:string, filename:string, size:number|null, sameOrigin:boolean, detectedAt:number, sourceElement:Element|null.
- **interface ParsedTorrent:** infohash:string, name:string, creationDate:number|null, comment:string|null, createdBy:string|null, pieceLength:number, isPrivate:boolean, trackers:readonly string[], source:string|null, files:readonly TorrentFileInfo[], totalSize:number, numPieces:number.
- **interface TorrentFileInfo:** path:readonly string[], length:number, fullPath:string.
- **type TorrentContentType** = "magnet" | "torrent-file".
- **interface DetectedTorrent** (unified detection type): id:string, type:TorrentContentType, magnet:MagnetInfo|null, torrentFile:TorrentFile|null, displayName:string, selected:boolean (mutable), sent:boolean (mutable), sendStatus:SendStatus|null (mutable), detectedAt:number.
- **type SendStatus** = "pending" | "sending" | "success" | "error" | "queued".
- **interface SendResult:** success:boolean, torrent:DetectedTorrent, error:string|null, response:Record<string,unknown>|null, completedAt:number.
- **interface PageScanResult:** pageUrl:string, pageTitle:string, items:readonly DetectedTorrent[], magnetCount:number, torrentFileCount:number, scannedAt:number, scanDurationMs:number.

src/types/config.ts

- **type AuthMethod** = "none"|"cookie"|"api_key"|"basic" (duplicated in types/api.ts).
- **interface ServerConfig** (28 fields): id, name, url, active:boolean, authMethod:AuthMethod, username:string|null, encryptedPassword:string|null, encryptedApiKey:string|null,

- requestTimeout:number, verifySsl:boolean,
- defaultCategory:string|null, defaultSavePath:string|null,
- startPaused:boolean, skipHashCheck:boolean,
- contentLayout:"original"|"subfolder"|"no_subfolder",
- autoTMM:boolean, uploadLimit:number (KiB/s, 0=unlimited),
- downloadLimit:number.
- **interface ExtensionConfig:** schemaVersion:number,
- servers:readonly ServerConfig[], activeServerId:string|null,
- autoScan:boolean, autoScanDelay:number,
- highlightTorrents:boolean,
- highlightStyle:"badge"|"border"|"glow",
- showNotifications:boolean, notificationSound:boolean,
- autoSend:boolean, maxHistoryItems:number, debugMode:boolean,
- healthCheckInterval:number (min), offlineQueue:boolean,
- maxOfflineQueueSize:number, showContextMenu:boolean,
- keyboardShortcuts:boolean, lastUpdated:number,
- encryptionKeyVersion:number.
- **const DEFAULT_CONFIG:** schemaVersion:1, servers:[],
- activeServerId:null, autoScan:true, autoScanDelay:2000,
- highlightTorrents:true, highlightStyle:"badge",
- showNotifications:true, notificationSound:false, autoSend:false,
- maxHistoryItems:100, debugMode:false, healthCheckInterval:5,
- offlineQueue:true, maxOfflineQueueSize:50, showContextMenu:true,
- keyboardShortcuts:true, lastUpdated:0, encryptionKeyVersion:1.
- **interface AutoDiscoveryConfig:** ports:readonly number[],
- scanTimeout:number, scanQbittorrent:boolean,
- scanFastApi:boolean, scanGo:boolean.
- **const DEFAULT_AUTO_DISCOVERY:** ports:[7187,7189,8080],
- scanTimeout:3000, all three scan flags true.
- **interface ConnectionTestResult:** success, url, version:string|
- null, error:string|null, responseTimeMs, testedAt.
- **interface ConfigChangeEvent:** key, newValue:unknown,
- oldValue:unknown, changedAt.

src/types/api.ts

qBitTorrent API + auth + health + Boba + queue + messaging
types: - **QBittorrentVersion:** version:string. -
QBittorrentAppPreferences: locale, save_path, temp_path,
preallocate_all:boolean, listen_port, max_active_downloads,
max_active_uploads, max_active_torrents, dl_limit, up_limit,
web_ui_username (all readonly). - **QBittorrentAddTorrentParams** (maps
/api/v2/torrents/add): urls?, torrents?:File, savepath?, cookie?,
category?, tags?, skip_checking? ("true"/"false" string), paused?,
root_folder?, rename?, upLimit?:number, dlLimit?:number, autoTMM?,
sequentialDownload?, firstLastPiecePrio?,
contentLayout?:"Original"|"Subfolder"|"NoSubfolder",
stopCondition?. - **QBittorrentAddResponse:** success:boolean,
error?:string. - **QBittorrentTorrentInfo:** hash, name, magnet_uri,
size, progress (0-1), dlspeed, upspeed, priority, num_seeds,
num_leechs, ratio, eta, state, category, tags, added_on,

completion_on, save_path. - **Auth credential interfaces:**
 CookieAuthCredentials{method:"cookie",username,password},
 ApiKeyAuthCredentials{method:"api_key",apiKey},
 BasicAuthCredentials{method:"basic",username,password},
 NoAuthCredentials{method:"none"}; union **AuthCredentials**. -
AuthState: method:AuthMethod, isAuthenticated:boolean,
 sidCookie:string|null, sidExpiresAt:number|null,
 basicAuthHeader:string|null, apiKeyHeader:string|null,
 lastRefreshedAt:number|null, consecutiveFailures:number. - **type**
AuthMethod = "none"|"cookie"|"api_key"|"basic". - **type HealthStatus**
 = "healthy"|"degraded"|"unhealthy"|"unknown". - **HealthCheckResult**:
 serverId, url, status:HealthStatus, version:string|null,
 responseTimeMs, authValid:boolean, error:string|null, checkedAt. -
BobaServerInfo: name, version, features:readonly string[],
 qbittorrent_connected:boolean, qbittorrent_version:string|null. -
BobaSearchResult: id, name, infohash, magnet_uri, size, seeders,
 leechers, source, upload_date, category. - **BobaSearchResponse**:
 results:readonly BobaSearchResult[], total, page, per_page. -
QueueItem: id, torrent:{infohash,magnetUri:string|
 null,torrentUrl:string|null,displayName}, serverId, addedAt,
 attempts, lastError:string|null, lastAttemptAt:number|null,
 priority:"high"|"normal"|"low". - **QueueProcessResult**: processed,
 succeeded, failed, remaining,
 results:ReadonlyArray<{itemId,success,error:string|null}>. - **type**
MessageType (20 values): scan-page, scan-result, send-torrent, send-
 result, get-detected, get-config, set-config, get-auth-state,
 authenticate, health-check, health-result, queue-process, queue-
 status, open-dashboard, show-notification, update-badge, torrent-
 detected, selection-change. - **ExtensionMessage**: type:MessageType,
 payload?:Record<string,unknown>, requestId?:string. -
ExtensionMessageResponse: success:boolean,
 data?:Record<string,unknown>, error?:string.

5. Shared-lib API surface (every exported fn/class signature)

src/shared/crypto.ts — AES-256-GCM credential encryption (Web Crypto)

- **interface EncryptedBundle**: salt:string (b64), iv:string (b64), ciphertext:string (b64, auth tag appended), version:number.
- async function encrypt(plaintext:string, passphrase:string): Promise<EncryptedBundle> — generates fresh random salt(16B) +IV(12B), derives key via PBKDF2, AES-GCM encrypt (tagLength 128). Throws StorageError on empty plaintext/passphrase or failure.
- async function decrypt(bundle:EncryptedBundle, passphrase:string): Promise<string> — re-derives key from

- salt+passphrase, AES-GCM decrypt. Throws `StorageError` (“passphrase may be incorrect”) on failure.
- function `generateSecurePassphrase(): string` — 32 random bytes → base64.
- function `isEncrypted(value:unknown): value is EncryptedBundle` — structural type guard.
- async function `sha256(input:string): Promise<string>` — hex SHA-256 digest.
- function `simpleHash(input:string): number` — non-crypto 32-bit hash (cache keys only).
- (private: `deriveKey`, `generateRandomBytes`, `bytesToBase64`, `base64ToBytes`.)

src/shared/errors.ts — error taxonomy

- **class `BobaLinkError` extends `Error`:** fields `code:string`, `statusCode:number|null`, `recoverable:boolean`, `cause:Error|null`, `context:Readonly<Record<string,unknown>>`. Ctor (message, options{code?,statusCode?,recoverable?,cause?,context?}). Methods `getUserMessage():string`, `toJSON():Record<string,unknown>`. Default code `BOBA_UNKNOWN`.
- **class `AuthError`** — code `BOBA_AUTH_FAILED`, `statusCode` default 401, `recoverable` true.
- **class `NetworkError`** — code `BOBA_NETWORK_ERROR`, `statusCode` null, `recoverable` true.
- **class `TorrentError`** — code `BOBA_TORRENT_ERROR`, `statusCode` default 400, `recoverable` true.
- **class `ParseError`** — code `BOBA_PARSE_ERROR`, `statusCode` null, `recoverable` **false**.
- **class `ConfigError`** — code `BOBA_CONFIG_ERROR`, `recoverable` true.
- **class `StorageError`** — code `BOBA_STORAGE_ERROR`, `recoverable` true.
- **class `RateLimitError`** — code `BOBA_RATE_LIMITED`, `statusCode` 429, `recoverable` true, extra field `retryAfter:number`. Ctor (message, `retryAfter`, options{context?}).
- **class `ServerError`** — code `BOBA_SERVER_<status>`, `statusCode` required, `recoverable` when `>=500 || ===429`. Ctor (message, `statusCode`, options).
- function `isBobaLinkError(value:unknown): value is BobaLinkError`.
- function `normalizeError(error:unknown, context={}): BobaLinkError`.

src/shared/events.ts — typed event emitter

- **interface `EventMap`** (13 events): `torrent-detected{id,type:"magnet"|"torrent-file",displayName,url}`, `scan-started{url,timestamp}`, `scan-completed{url,magnetCount,torrentFileCount,durationMs}`, `scan-error{url,error}`, `send-started{ids}`, `send-completed{results[{id,success}]}`, `send-error{id,error}`, `auth-`

```
state-changed{method,authenticated}, connection-
status{serverId,connected,latency:number|null}, config-
changed{key,newValue,oldValue}, queue-updated{size}, badge-
update{count,color},
notification{title,message,type:"info"|"success"|"warning"|"err
or"}.
```

- **type** EventName = keyof EventMap; **type** EventListener<T> = (payload:EventMap[T])=>void.
- **class** TypedEventEmitter: on<T>(event,listener):()=>void (returns unsub); once<T>(event,listener):void; emit<T>(event,payload):void (try/catch per listener); off(event?):void; listenerCount(event):number; hasListeners(event):boolean.
- **const** globalEvents = new TypedEventEmitter().

src/shared/logger.ts — structured logging

- type LogLevel = "debug"|"info"|"warn"|"error" (severity 0/1/2/3). Default min level info; debug when debug mode on.
- function initLogger(isDebug:boolean): void.
- function debug/info/warn/error(context:string, message:string, data?:unknown): void (error always emitted).
- function timed(context, operation): ()=>void — logs start, returns end fn logging duration.
- function createLogger(context:string) → {debug,info,warn,error,timed} (bound). **type** Logger = ReturnType.
- function isDebugEnabled(): boolean.
- Format: [ISO-timestamp] [LEVEL] [context] message.

src/shared/storage.ts — chrome.storage.local wrapper

- async function storageGet<T>(key): Promise<T|null>.
- async function storageSet<T>(key, value): Promise<void>.
- async function storageRemove(key): Promise<void>.
- async function storageGetMultiple<T>(keys:readonly string[]): Promise<ReadonlyMap<string,T|null>>.
- async function storageClearAll(): Promise<void> — removes only bobalink_-prefixed keys.
- function getStorageKeys(): typeof STORAGE_KEYS.
- function onStorageChange<T>(keys, callback): ()=>void — local-area only; returns unsub.
- **class** NamespacedStorage: ctor (namespace) → prefix bobalink_<namespace>; methods get<T>(key), set<T>(key,value), remove(key), getAllKeys(), clear(). All errors wrapped as StorageError.

src/shared/utils.ts — utilities

- function debounce<T extends unknown[]>(fn, delay): {(...args:T):void; cancel():void}.
 - function throttle<T extends unknown[]>(fn, interval): {(...args:T):void; cancel():void}.
 - function generateId(): string — crypto.randomUUID or timestamp+random fallback.
 - function truncate(str, maxLength): string — ellipsis.
 - function escapeHtml(text): string — via textContent/innerHTML (XSS-safe; **DOM-dependent**, not service-worker safe).
 - function sleep(ms): Promise<void>.
 - async function retryWithBackoff<T>(fn, maxRetries=3, baseDelay=1000, maxDelay=30000): Promise<T> — exponential backoff + 0.3 jitter.
 - **class TokenBucket**: ctor (capacity, refillRate); consume():boolean; getAvailableTokens():number; private refill().
 - function yieldToBrowser(): Promise<void> — rAF or setTimeout(0).
 - async function processInChunks<T>(items, processor, chunkSize=50): Promise<void>.
 - function isValidHttpUrl(url): boolean.
 - function getDomain(url): string — hostname or "".
 - function formatBytes(bytes, decimals=2): string — B/KB/MB/GB/TB/PB.
 - function deepClone<T>(obj): T — JSON round-trip.
 - function arraysEqual<T>(a, b): boolean — shallow.
-

6. Every constant value (src/shared/constants.ts) — esp. URLs/ports/timeouts/secrets

Regex patterns

- MAGNET_REGEX = /magnet:\?xt=urn:btih:[a-fA-F0-9]{40}[\^s"'<>]*\$/gi
- MAGNET_VALIDATION_REGEX = /^magnet:\?xt=urn:btih:[a-fA-F0-9]{40}(\?:[&][\^s"'<>]*)?\$/i
- INFOHASH_REGEX = /xt=urn:btih:([a-fA-F0-9]{40})/i
- TORRENT_FILE_REGEX = /https?:\//[\^s"'<>]+\\.torrent(\?:\?[\^s"'<>]*)?/gi
- TORRENT_FILE_VALIDATION_REGEX = /^https?:\//.+\.torrent(\?.*)?\$/i
- INFOHASH_HEX_REGEX = /^[a-fA-F0-9]{40}\$/
- INFOHASH_BASE32_REGEX = /^[A-Z2-7]{32}\$/
- MAGNET_DN_REGEX = /[?&]dn=([\^&]*)/

- `MAGNET_TR_REGEX = /[?&]tr=(^[^&]*)/g`

Ports / URLs / endpoints (load-bearing)

- `DEFAULT_PORTS = {FAST_API:7187, GO:7189, QBITTORRENT:8080}`
- `DEFAULT_URLS = {FAST_API:"http://localhost:7187", GO:"http://localhost:7189", QBITTORRENT:"http://localhost:8080"}`
- `QBITTORRENT_ENDPOINTS:`
 - `AUTH_LOGIN: "/api/v2/auth/login", AUTH_LOGOUT: "/api/v2/auth/logout"`
 - `APP_VERSION: "/api/v2/app/version", APP_PREFERENCES: "/api/v2/app/preferences", APP_SET_PREFERENCES: "/api/v2/app/setPreferences"`
 - `TORRENTS_ADD: "/api/v2/torrents/add", TORRENTS_INFO: "/api/v2/torrents/info", TORRENTS_DELETE: "/api/v2/torrents/delete", TORRENTS_PAUSE: "/api/v2/torrents/pause", TORRENTS_RESUME: "/api/v2/torrents/resume"`
 - `TRANSFER_INFO: "/api/v2/transfer/info"`

Port reconciliation needed: extension assumes qBitTorrent direct on **8080**; Boba's actual proxy exposes qBitTorrent WebUI on **7186** (merge service 7187, boba-jackett 7189). FastAPI 7187 + Go 7189 match. Magnet/.torrent send goes through `/api/v2/torrents/add` on the chosen base URL.

Timing / retry / rate limit

- `DEBOUNCE_DELAYS = {MUTATION:500, STORAGE_WRITE:250, AUTO_SCAN:1000, BADGE_UPDATE:300, NOTIFICATION:2000} (ms)`
- `REQUEST_TIMEOUTS = {DEFAULT:15000, HEALTH_CHECK:5000, AUTH:10000, ADD_TORRENT:30000, AUTO_DISCOVERY:3000} (ms)`
- `RETRY_CONFIG = {MAX_RETRIES:3, BASE_DELAY_MS:1000, MAX_DELAY_MS:30000, JITTER_FACTOR:0.3}`
- `RATE_LIMIT = {MAX_REQUESTS:10, WINDOW_MS:1000} (note: differs from DB seed behavior.rate_limit_requests=30)`

Storage keys (STORAGE_KEYS, all bobalink_-prefixed)

- `CONFIG:"bobalink_config", AUTH_STATE:"bobalink_auth_state", CREDENTIALS:"bobalink_credentials", DETECTED:"bobalink_detected", HISTORY:"bobalink_history", QUEUE:"bobalink_queue", HEALTH:"bobalink_health", KEY_MATERIAL:"bobalink_key_material"`

Encryption (ENCRYPTION) — §11.4.10 secret-handling scheme

- ALGORITHM: "AES-GCM", KEY_LENGTH_BITS: 256, IV_LENGTH_BYTES: 12, SALT_LENGTH_BYTES: 16, KDF_ALGORITHM: "PBKDF2", KDF_ITERATIONS: 100000, KDF_HASH: "SHA-256", CURRENT_KEY_VERSION: 1.
- Scheme: passphrase → PBKDF2 (100k iters, SHA-256, 16-byte random salt) → AES-256-GCM (12-byte random IV, 128-bit auth tag). Each encrypt() is independent (fresh salt+IV). Stored bundle = {salt, iv, ciphertext, version} all base64. Aligns with Boba's AES-256-GCM master-key model (boba-jackett). The passphrase source/storage is **not defined in these foundation files** — see open questions.

UI constants

- BADGE_COLORS = {HEALTHY: "#4CAF50", DEGRADED: "#FF9800", ERROR: "#F44336", SCANNING: "#2196F3", DETECTED: "#9C27B0", DEFAULT: "#757575"}
- ICON_SIZES = [16, 32, 48, 128]

Site selectors (SITE_SELECTORS, code-level supplement to DB table)

Record<domain, selectors[]>: generic (a[href^="magnet:"], a[href\$=".torrent"], a[href*=".torrent?"], a[href*="download.php?id="]); plus per-domain entries for 1337x.to, thepiratebay.org/theiratebay10.org, rarbg.to/rarbgtorrents.org, yts.mx/yts.lt, eztv.re, limetorrents.lol, torrentgalaxy.to, nyaa.si, animetosho.org, torrentz2.eu, fitgirl-repacks.site, rutracker.org, katcr.co, demonoid.is, iptorrents.com (a[href\$=".torrent"], a[href*="download.php/"]), torrentleech.org (a[href\$=".torrent"], a[href*="/download/"]), beyond-hd.me, passthepopcorn.me.

Misc (EXT)

- ID: "bobaLink", NAME: "BobaLink", DEFAULT_CATEGORY: "BobaLink", MAX_POPUP_ITEMS: 50, MAX_DISPLAY_NAME_LENGTH: 80.
-

7. CI flag + which build/test commands to keep as manual scripts

CONSTITUTION FLAG (Boba forbids ALL CI/CD — Hard Stop rule, CLAUDE.md): the reference repo ships **two GitHub Actions workflows** that **MUST NOT** be ported:

- `.github/workflows/ci.yml` — jobs: lint (lint + format:check + compile), unit-test (test + coverage upload), build (npm run build + dist upload). Triggers on push/PR to main/develop.
- `.github/workflows/release.yml` — 6 jobs: version-bump (semver validate, CHANGELOG check, manifest version edit), build-all (matrix chrome/firefox), sign-firefox (AMO via web-ext sign + Mozilla secrets), create-release (GitHub Release), upload-chrome (Chrome Web Store publish API), release-summary. Uses secrets MOZILLA_API_KEY/SECRET, FIREFOX_EXTENSION_ID, CHROME_CLIENT_ID/SECRET/REFRESH_TOKEN, CHROME_WEBSTORE_ID.

Both .yml files MUST be deleted / never copied into the Boba tree (no `.github/workflows/*.yml` allowed; CI is manual via `./ci.sh`). Extract only the **commands** worth keeping as manual scripts:

Keep as manual local commands (e.g. wired into a `ci.sh`-style script or Makefile target): - **Lint/format/compile gate:** npm run lint → npm run format:check → npm run compile (tsc --noEmit). - **Unit tests + coverage:** npm run test (jest --coverage, 80% threshold). - **E2E:** npm run test:e2e (playwright test) — requires built dist/ + globalSetup/teardown files (absent; must be authored). - **Build:** npm run build (chrome) / npm run build:firefox; package: npm run zip / zip:firefox. - **Release packaging mechanics worth reusing manually** (NOT as automation): zip dist dir, sha256sum checksum, inject VERSION/BUILD_SHA/BUILD_DATE files into dist. Firefox web-ext sign and Chrome Web Store upload are out of scope for the local-manual model (and need real store credentials — handle per §11.4.10 if ever used, never in-repo). - The release workflow's `sed -i "s/1.0.0/$VERSION/g" src/sql/schemas.sql src/sql/migrations/001_initial.sql` shows version is also embedded in SQL seed rows — a manual bump script must touch package.json + wxt.config.ts manifest version + both SQL files + app_metadata/extension_config schema_version/app_version rows together.

8. Open questions / drift to resolve in the plan

1. **Port mismatch (critical).** Extension hardcodes qBitTorrent on **8080** (host_permissions, CSP, DEFAULT_URLS/PORTS, DB defaults). Boba's real qBitTorrent WebUI is proxied on **7186** (CLAUDE.md port table); 7187=merge service, 7189=boba-

jackett. FastAPI(7187)/Go(7189) match but qBitTorrent must change 8080→7186 across wxt.config.ts (host_permissions + CSP connect-src), constants.ts, SQL seeds, and DEFAULT_AUTO_DISCOVERY.ports.

2. **Encryption passphrase provenance undefined.** crypto.ts requires a passphrase for encrypt/decrypt and STORAGE_KEYS.KEY_MATERIAL / EXTENSION/encryptionKeyVersion exist, but **where the passphrase comes from / how it's persisted** is not in the foundation layer (must be in background/options — not in scope here). Must confirm it satisfies §11.4.10 (no plaintext secret at rest, no secret in git). The server.api_key/server.password SQL seeds are empty strings (data_type=encrypted) — fine.
3. **Dual storage model.** Two parallel persistence designs exist: (a) the SQL schema (sql.js over chrome.storage) with 9 tables, and (b) the STORAGE_KEYS JSON-blob model + NamespacedStorage. The foundation does not show which is authoritative at runtime (storage.ts only implements the JSON-blob path; no sql.js wrapper file present in this layer). Plan must decide one source of truth.
4. **Type duplication.** AuthMethod defined identically in both types/api.ts and types/config.ts; ServerConfig (TS) and server_config (SQL) diverge in field naming (url vs base_url, encryptedPassword vs password_encrypted, port fields only in SQL). Reconcile.
5. **Config-value mismatch.** RATE_LIMIT.MAX_REQUESTS=10/WINDOW_MS=1000 (constants) vs DB seed behavior.rate_limit_requests=30. And autoScanDelay default 2000 (config) vs DEBOUNCE_DELAYS.AUTO_SCAN=1000. Decide canonical.
6. **Referenced-but-absent files** (in this layer): tests/e2e/global-setup.ts, tests/e2e/global-teardown.ts, src/assets/icon.svg, popup/options/content/background source, and the sql.js wrapper. These are downstream layers, not foundation — flagged so the plan doesn't assume they exist yet.
7. **Release workflow references non-existent npm scripts** (build:chrome) and non-existent src/manifest.chrome.json/manifest.firefox.json (WXT generates the manifest, there are no static manifest files). Drift in the reference repo — ignore for porting.
8. **Repo URL is HTTPS** (<https://github.com/boba-project/bobalink.git>) — Boba constitution mandates SSH remotes (git@...). Fix when integrating.
9. **minimum_chrome_version "109"** + MV3 module service worker — confirm target browser floor is acceptable; Firefox MV3 support (release.yml builds firefox) needs verification of background type:module SW support.